SCOPE: Selective Critical-Step On-Policy Self-Distillation for Agentic Reinforcement Learning

Anonymous Author(s)

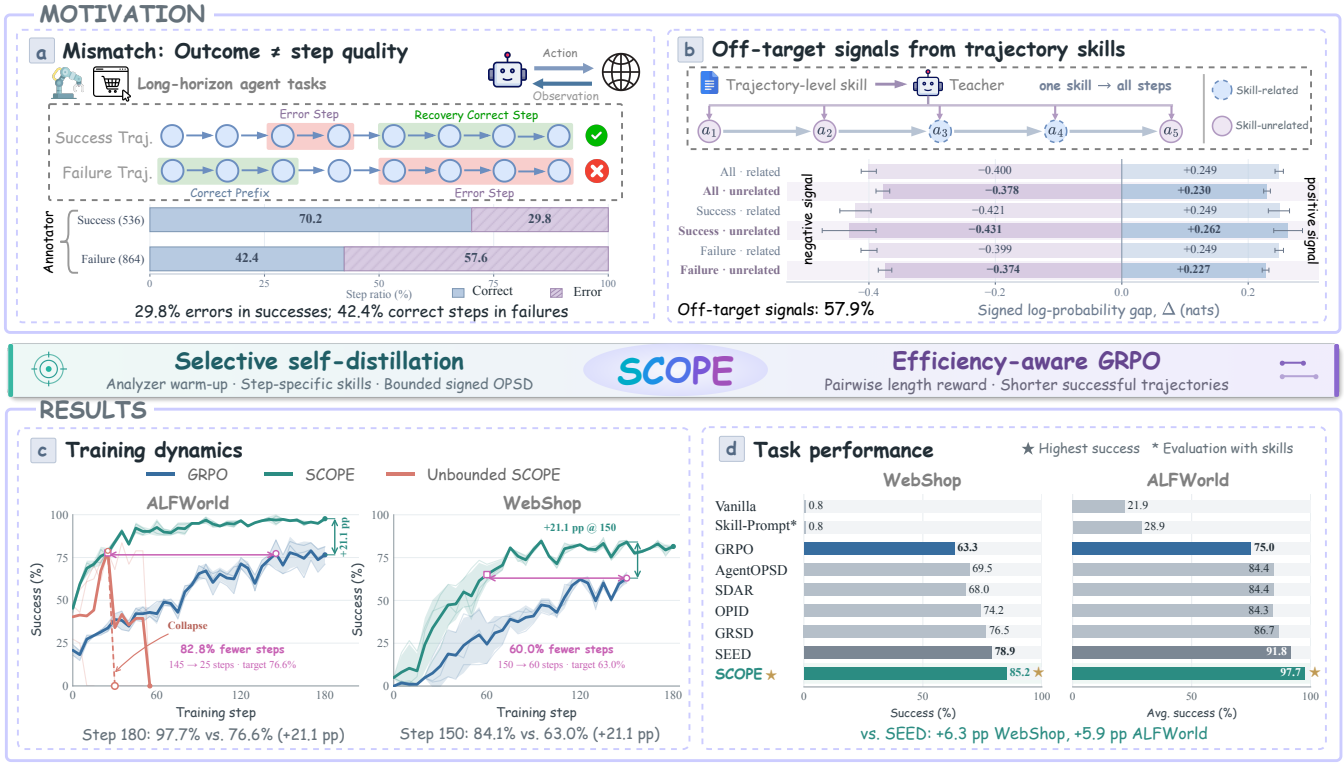


Figure 1: Motivation and results of SCOPE. (a) Outcome–step mismatch. (b) Off-target skill signals with 95% bootstrap intervals. (c) Training dynamics across multiple seeds, with shaded bands indicating ±1 standard deviation. "Unbounded SCOPE" uses raw-gap OPSD; step savings are measured relative to GRPO's final success rate. (d) Qwen2.5-3B main results at step 150.

Abstract

Outcome-based reinforcement learning provides coarse supervision for long-horizon LLM agents: failed trajectories contain correct steps, while successful trajectories can contain errors. Trajectory-level skills can also shift teacher preferences at unrelated steps, introducing off-target distillation signals. We introduce **SCOPE**, a framework for *Selective Critical-Step On-Policy Self-Distillation*. After a supervised warm-up, the policy analyzes its own trajectories, generates step-specific hindsight skills, and distills their guidance only at selected error steps. A bounded signed teacher–student coefficient stabilizes auxiliary updates while preserving positive reinforcement and negative correction. A success-relative length reward further encourages efficient task completion. Across three backbones, SCOPE achieves the highest WebShop success and the highest or tied highest ALFWorld success among the evaluated methods. With Qwen2.5-3B, it reaches 97.7% on ALFWorld and 85.2% on WebShop, improving over SEED by 5.9 and 6.3 percentage points, respectively. On unseen ALFWorld tasks, SCOPE reaches 89.6% success versus 66.4% for GRPO.

1 Introduction

LLM agents solve tasks through sequences of tool calls and environment interactions [13, 20, 37]. Reinforcement learning (RL) improves these decisions using task feedback [9, 18, 27]. In particular, GRPO [22] converts the outcomes of sampled trajectories into relative advantages, providing a scalable training signal without step-level annotations [16, 27, 30]. However, an outcome reward reveals whether a task was completed, with little indication of which intermediate decisions were sound or required correction.

This distinction matters because **trajectory success and step quality do not always agree**. A successful agent may recover from invalid or mistaken actions, while a failed agent may make useful progress before a consequential error. We examine this *outcome–step mismatch* in annotated WebShop trajectories. Figure 1(a) shows 29.8% erroneous steps in the successful group and 42.4% correct

,
2026.

steps in the failed group. These observations motivate identifying errors independently of the final outcome. When an outcome-level advantage is shared across a trajectory, it cannot distinguish these local differences in decision quality [5, 6, 11].

On-policy self-distillation (OPSD) [42] offers a way to supplement task rewards with local guidance: a teacher branch receives privileged context and scores actions sampled by the ordinary policy [1, 12]. Trajectory-derived skills summarize useful hindsight [3], and methods such as SEED train the policy to generate this guidance from its own experience [26, 29]. However, **privileged skill context can induce off-target probability shifts at steps unrelated to the skill**. A trajectory-level skill may address a particular mistake or subgoal, yet conditioning the teacher on it throughout the trajectory can also change token probabilities at other steps. EOPSA [12] highlights a related issue: privileged prompts can introduce task-irrelevant discrepancies that dilute useful distillation signals. In our agentic setting, Figure 1(b) shows that skill-unrelated steps account for 57.9% of nonzero sampled-token gaps. Among these off-target signals, 53.9% are negative and 46.1% are positive. Distilling across all steps assigns auxiliary updates to these probability shifts even where the skill offers no correction for the current decision. The second problem is therefore the propagation of privileged-context effects beyond the steps where the skill provides corrective guidance.

Hierarchical hindsight skills and group reflection offer finer-grained ways to use trajectory experience [23, 35, 43]. Building on these advances, we focus on matching the scope of each corrective skill to its target decisions. The two problems motivate a common design principle: *identify errors independently of the trajectory outcome, and apply corrective guidance only where it is relevant.*

We introduce **SCOPE**, a framework for *Selective Critical-Step On-Policy Self-Distillation*. Following a lightweight supervised warm-up, the policy analyzes its own trajectories to identify critical errors in both successful and failed episodes and generates corresponding step-specific skills. A skill-conditioned self-teacher then provides corrective supervision exclusively at the selected steps, aligning distillation with the decisions that require correction.

To stabilize optimization, SCOPE employs a *bounded signed* distillation coefficient that limits extreme teacher–student discrepancies while preserving both reinforcement and suppression. Furthermore, a success-relative pairwise length reward encourages efficient task completion. Together, these designs enable stable, targeted, and efficiency-aware agentic RL without additional guidance at inference time.

Experiments on ALFWorld and WebShop across three backbones demonstrate consistent improvements. With Qwen2.5-3B, SCOPE achieves 97.7% and 85.2% success, outperforming SEED by 5.9 and 6.3 percentage points, respectively. On unseen ALFWorld tasks, SCOPE improves success from 66.4% to 89.6% while reducing average interactions by 46.7%.

Our contributions are:

- We identify two limitations of trajectory-level supervision: outcome–step mismatch and off-target distillation signals induced by privileged skill context at unrelated steps.
- We develop SCOPE, which internalizes error analysis and applies self-generated corrective supervision at selected critical steps.
- We combine bounded signed distillation for stable corrective updates with success-relative length rewards for efficient task completion. Experiments across three backbones and an unseen task split assess task success, training dynamics, and interaction efficiency.

2 Related Work

2.1 Reinforcement Learning for LLM Agents

Reinforcement learning trains LLM agents through feedback from tool use and environment interaction [15, 31, 32]. ToolRL [18], RAGEN [27], and Search-R1 [9] apply this approach to tool selection, interactive planning, and information seeking. GRPO [22] provides a common optimization backbone by comparing trajectory outcomes within a sampled group. However, a shared outcome advantage offers limited resolution over the intermediate decisions of a long interaction. GiGPO [6] estimates step-level advantages through hierarchical grouping, while GraphGPO [5] uses state-transition graphs to refine credit assignment. SCOPE supplements outcome-based RL with hindsight error localization and corrective guidance at the decisions that require it.

2.2 Hindsight Skills for Agents

Experience can guide agents through verbal reflection, reusable skills, or explicit memory, as demonstrated by Reflexion [23], Voyager [25], and ExpeL [41]. Recent methods internalize such guidance through policy training [17, 26, 28, 35]. SDAR [14] retrieves skills from an external skill bank as privileged teacher context and combines GRPO with a sigmoid-gated auxiliary distillation objective. Its guidance depends on retrieval quality and skill applicability: retrieved skills can be irrelevant, incomplete, or redundant, producing unreliable teacher preferences. The gate therefore emphasizes positive teacher endorsements and attenuates negative rejections.

SEED [29] replaces skill retrieval with a policy-synchronized hindsight analyzer. A supervised warm-up teaches the model to generate trajectory-level skills. During RL, the policy serves as both actor and analyzer, enabling self-evolving guidance. However, trajectory-level skills condition distillation across all steps, potentially introducing off-target signals at unrelated decisions (Figure 1(b)). OPID [35] uses an external analyzer to construct hierarchical skills, retaining episode-level guidance at non-critical steps. GRSD [43] derives group-level guidance from policy-generated reflections evaluated by an external judge, refining turn-level credit assignment. In contrast, SCOPE internalizes error analysis and restricts skill-conditioned distillation to selected error steps in both successful and failed trajectories.

2.3 On-Policy Self-Distillation

On-policy distillation supervises learner-generated sequences, reducing the distribution mismatch associated with offline teacher demonstrations [1, 7, 10, 19]. Self-distillation constructs a teacher from the same model under privileged context [12, 42]. Skill-SD [26] uses trajectory-derived skills for multi-turn distillation. RLSD [34] uses teacher–student differences to modulate token-level RL updates, while SDAR [14] bounds the strength of an auxiliary distillation objective. AgentOPSD [28] aggregates token-level discrepancies and recursively updates belief states for turn-level credit

assignment. SOD [44] adjusts distillation strength according to step-level divergence.

However, teacher–student discrepancies do not necessarily reflect task-relevant corrections. EOPSA [12] highlights how privileged prompting can introduce task-irrelevant stylistic shifts that dilute useful distillation signals. In agentic tasks, skill conditioning may similarly affect decisions unrelated to the intended correction. SCOPE addresses this issue through step-specific skills and selective error-step distillation, while a bounded signed coefficient controls update strength and preserves both positive reinforcement and negative correction.

3 Method

SCOPE supplements outcome-based RL with corrective supervision at selected error steps (Figure 2). A supervised warm-up teaches the policy to locate mistakes and generate corrective skills. During online training, the same policy analyzes its rollouts, and a skill-conditioned self-teacher scores the sampled actions at the selected steps. Bounded signed self-distillation transfers these corrections to the student, while a success-relative length reward encourages efficient task completion.

3.1 Agent Interaction and Policy Snapshots

For a task q, the agent receives observation o_t and generates a response $a_t = (y_{t,1}, \ldots, y_{t,L_t})$ from its available interaction history h_t:

$$h_t = (q, o_1, a_1, \ldots, o_t), \qquad a_t \sim \pi_\theta(\cdot \mid h_t). \tag{1}$$

The response contains the model-generated reasoning and action tokens for one environment interaction. The environment executes the action and returns o_{t+1}. A completed trajectory $\tau_i = \{(h_{i,t}, a_{i,t}, o_{i,t+1})\}_{t=1}^{T_i}$ contains T_i interaction steps and receives a task reward R_i. We denote its binary success indicator by $z_i \in \{0, 1\}$, where $z_i = 1$ indicates successful task completion. Interaction length T_i and response-token length $L_{i,t}$ are distinct quantities.

At outer iteration u, we take a policy snapshot $\bar{\theta}_u = \theta_u$ and sample N trajectories for each task:

$$\mathcal{G}_q^{(u)} = \{\tau_i\}_{i=1}^{N}, \qquad \tau_i \sim \pi_{\bar{\theta}_u}(\cdot \mid q). \tag{2}$$

This snapshot supplies the actor, the hindsight analyzer, and the privileged teacher used to construct the update data. The trainable student starts from this checkpoint and updates its parameters θ across successive minibatches, while $\bar{\theta}_u$ remains fixed for the current outer iteration.

3.2 Self-Bootstrapped Error Analysis

Supervised initialization. We first train the policy to analyze trajectories through supervised fine-tuning, following SEED [29]. GLM-5.2 [39] labels errors in a small pool of base-policy trajectories and writes a short corrective skill for each error or group of related errors. A skill entry is represented as $(\mathcal{D}_{i,k}, s_{i,k}, e_{i,k})$, where $\mathcal{D}_{i,k}$ is a nonempty set of target-step indices, $s_{i,k}$ is the corrective skill, and $e_{i,k}$ contains the error explanation and supporting evidence. A singleton target set describes one mistake; several steps may share a skill when they require the same correction.

Let x_i be the serialized analysis input and Y_i its structured annotation. The warm-up minimizes the autoregressive objective

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\mathbb{E}_{(x_i, Y_i)\sim\mathcal{D}_{\mathrm{SFT}}}\left[\sum_{r=1}^{|Y_i|} \log \pi_\theta(Y_{i,r} \mid x_i, Y_{i,<r})\right], \tag{3}$$

where $Y_{i,r}$ is an annotation token. The resulting checkpoint initializes online training. External annotation is confined to this offline stage. Appendix B describes data preparation, and Appendix C gives the SFT and analyzer prompt templates.

Online hindsight analysis. The analyzer inspects successful and failed trajectories separately. For a successful trajectory, it uses the trajectory's own observations, responses, and action-validity feedback to identify avoidable mistakes. For a failed trajectory, it additionally receives the action sequences of successful trajectories from the same on-policy task group. With $\mathrm{Act}(\tau)$ denoting an extracted action sequence, the reference context is

$$\mathcal{R}_i = \begin{cases} \{\mathrm{Act}(\tau_j) : \tau_j \in \mathcal{G}_q^{(u)},\ z_j = 1\}, & z_i = 0, \\ \varnothing, & z_i = 1. \end{cases} \tag{4}$$

Successful references demonstrate feasible task progress. The analyzer judges each decision using the information available at that step and the resulting environmental feedback, allowing valid alternative strategies and reasonable exploration.

The policy snapshot generates the analysis under an analysis-specific prompt:

$$\begin{aligned} &\widehat{Y}_i \sim \mathcal{A}_{\bar{\theta}_u}(\cdot \mid \tau_i, \mathcal{R}_i), \\ &\widehat{Y}_i = \{(\mathcal{D}_{i,k}, \widehat{s}_{i,k}, \widehat{e}_{i,k})\}_{k=1}^{J_i}, \qquad J_i \le K_{\max}. \end{aligned} \tag{5}$$

The limit $K_{\max}$ bounds the number of skill entries, not the total number of selected steps. Parsing removes nonexistent target indices and duplicate assignments, so the retained target sets are disjoint. The selected-step set is

$$C_i = \bigcup_{k=1}^{J_i} \mathcal{D}_{i,k}, \qquad m_{i,t} = \mathbb{I}[t \in C_i]. \tag{6}$$

Each selected step receives the skill of its assigned entry, denoted $\widehat{s}_{i,t}$. The analyzer may return an empty set when no clear error is supported, including when a failed group supplies no successful reference. An unselected step receives no auxiliary distillation signal.

At the next outer iteration, the updated policy supplies both the actor and analyzer.

3.3 Bounded Signed Self-Distillation

Step-specific teacher guidance. We use skill-conditioned on-policy self-distillation to transfer hindsight guidance into the ordinary policy [26, 35]. For each selected step, a context augmentation function H inserts its assigned skill into the ordinary policy input: $\widetilde{h}_{i,t} = H(h_{i,t}, \widehat{s}_{i,t})$. We retain the original sampled response and its within-response prefix. Teacher and student score the same token, using parameters $\vartheta_T = \bar{\theta}_u$ and $\vartheta_S = \theta$ and contexts $h_{i,t}^T = \widetilde{h}_{i,t}$ and $h_{i,t}^S = h_{i,t}$:

$$\ell_{i,t,\ell}^b = \log \pi_{\vartheta_b}(y_{i,t,\ell} \mid h_{i,t}^b, y_{i,t,<\ell}), \quad b \in \{T, S\}. \tag{7}$$

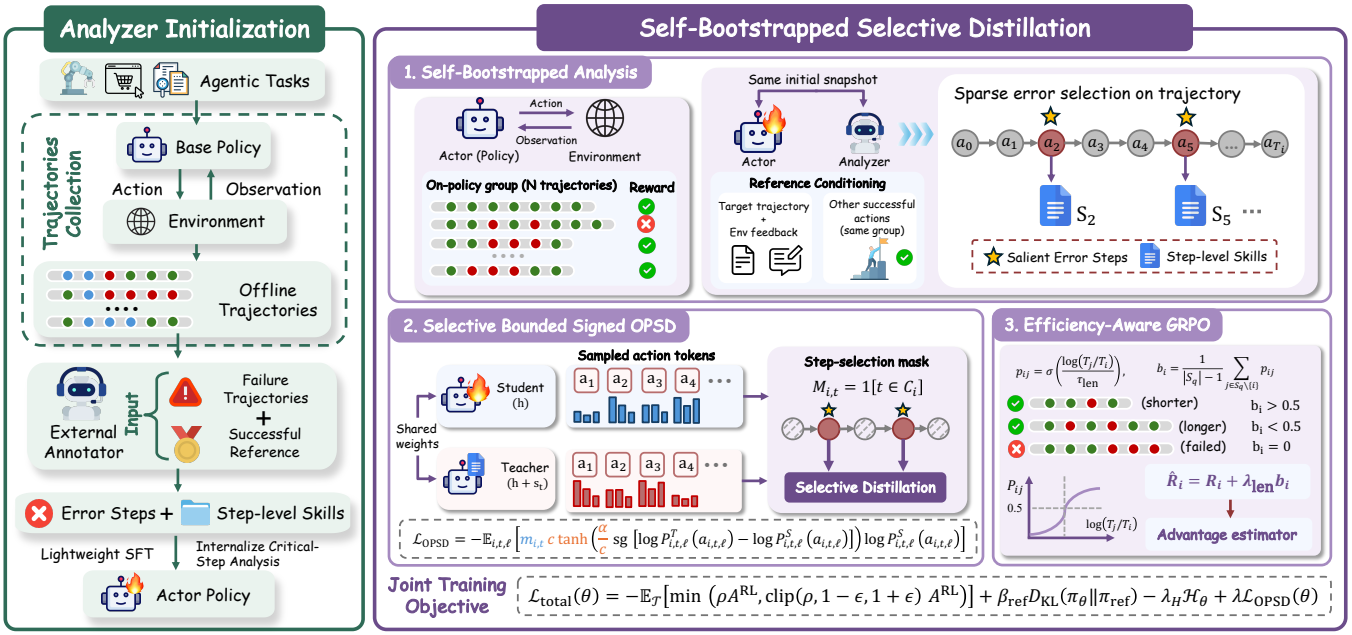


Figure 2: Overview of SCOPE. Supervised initialization teaches the policy to analyze trajectories and generate corrective skills. During online training, the policy collects and analyzes its own rollouts, assigns skills to selected error steps, and combines selective self-distillation with efficiency-aware GRPO.

Teacher log-probabilities are computed once before actor optimization and cached for the selected steps. Student log-probabilities are recomputed at each training forward pass as θ changes.

Bounding the signed learning signal. Figure 3 shows an anomaly during minibatch optimization. At outer update 26, the pre-optimization gap is 0.284 nats. During training, the student's mean sampled-token log-probability drops to −2716.0 nats, while the cached teacher mean remains at −0.440 nats. The training-stage gap reaches 2715.5 nats, with a non-finite recorded gradient norm.

Teacher scores are cached before actor optimization. Successive minibatch updates change θ, so training forwards compare updated student scores with the fixed teacher cache. A sharp drop in student probabilities on sampled tokens can therefore turn a moderate pre-optimization gap into an extreme training-stage discrepancy. Ordinary sampled-token OPSD uses this gap to weight the student's score gradient, amplifying auxiliary updates.

We control gap-driven amplification by replacing the raw gap with a bounded signed coefficient:

$$\Delta_{i,t,\ell}(\theta) = \mathrm{sg}[\ell^{T}_{i,t,\ell} - \ell^{S}_{i,t,\ell}(\theta)],$$
$$\widehat{A}^{\mathrm{dist}}_{i,t,\ell} = c\,\tanh\!\left(\frac{\alpha\Delta_{i,t,\ell}}{c}\right), \tag{8}$$

where sg stops gradients through the coefficient. The parameters $\alpha > 0$ and $c > 0$ control the slope at zero and saturation magnitude: $\widehat{A}^{\mathrm{dist}} \approx \alpha\Delta$ for small gaps, and $|\widehat{A}^{\mathrm{dist}}| < c$ for finite gaps. This symmetric transformation preserves the gap's sign and gives zero guidance when teacher and student agree on the sampled token.

Let $\mathcal{T}$ denote the valid, retained response-token positions in the current training microbatch, and let $\mathcal{T}_{\mathrm{sel}}$ contain only those

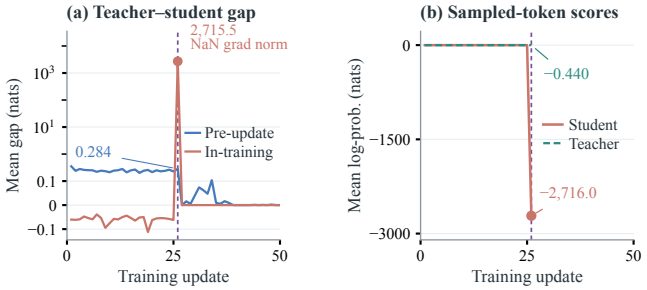


Figure 3: Anomaly during minibatch optimization. (a) Pre-optimization and training-stage gaps (symmetric-log axis). (b) Current student and cached teacher sampled-token log-probabilities. Dashed lines mark outer update 26.

positions belonging to selected error steps. Writing $N_{\mathrm{sel}} = |\mathcal{T}_{\mathrm{sel}}|$, the distillation loss is simply the mean over these selected tokens:

$$\mathcal{L}_{\mathrm{OPSD}}(\theta) = -\frac{c}{N_{\mathrm{sel}}} \sum_{(i,t,\ell)\in\mathcal{T}} m_{i,t} \cdot \tanh\!\left(\frac{\alpha}{c}\,\mathrm{sg}[\ell^{T}_{i,t,\ell} - \ell^{S}_{i,t,\ell}(\theta)]\right) \ell^{S}_{i,t,\ell}(\theta). \tag{9}$$

Unselected steps do not enter this loss or its token count. If $N_{\mathrm{sel}} = 0$, we set $\mathcal{L}_{\mathrm{OPSD}} = 0$.

For each selected token, the contribution to the policy gradient is $-\widehat{A}^{\mathrm{dist}}\nabla_\theta \ell^{S}/N_{\mathrm{sel}}$. The derivative with respect to its student log-probability has magnitude at most c/N_{sel}. With $c = 1$, the coefficient is confined to $[-1, 1]$, controlling the influence of extreme teacher–student discrepancies on distillation updates. The signed transformation reinforces positive-gap tokens and suppresses negative-gap tokens, retaining corrective guidance in both directions.

SDAR [14] also addresses OPSD instability with a bounded sigmoid gate. Its nonnegative auxiliary coefficient attenuates teacher rejections without retaining their suppressive direction. SCOPE preserves this negative guidance at selected error steps through a signed coefficient, while bounding both update directions. Appendix A.1 derives the fixed-data log-cosh interpretation and auxiliary-gradient bounds; Appendix A.2 bounds the contribution of irrelevant selected tokens.

3.4 Success-Relative Pairwise Efficiency

Selective distillation corrects identified errors. Successful trajectories can also contain redundant interactions. Length-aware rewards have been explored to balance correctness and computational cost in reasoning models. L1 [2] combines correctness with adherence to a specified token-length budget, while ShorterBetter [38] uses the shortest correct response in a sampled group as a dynamic length reference. For multi-turn agents, we measure efficiency by the number of environment interactions and compare successful trajectories for the same task within the current rollout group. This provides a policy-relative reference as successful interaction lengths evolve during training.

Let $\mathcal{S}_q = \{i : \tau_i \in \mathcal{G}_q^{(u)},\ z_i = 1\}$. For two distinct successful trajectories, we use a Bradley–Terry pairwise comparison [4] with utility $-\log T_i/\tau_{\text{len}}$:

$$p_{ij} = \sigma\left(\frac{\log T_j - \log T_i}{\tau_{\text{len}}}\right), \tag{10}$$

where $\sigma(x) = (1 + e^{-x})^{-1}$ and $\tau_{\text{len}} > 0$ controls the smoothness of the preference. The score exceeds $1/2$ when trajectory i is shorter than trajectory j. Averaging over the other successful trajectories gives

$$b_i = \begin{cases} \dfrac{1}{|\mathcal{S}_q| - 1} \displaystyle\sum_{j \in \mathcal{S}_q \setminus \{i\}} p_{ij}, & |\mathcal{S}_q| \geq 2, \\ \frac{1}{2}, & |\mathcal{S}_q| = 1, \end{cases} \qquad i \in \mathcal{S}_q. \tag{11}$$

Each complete trajectory enters this comparison once, regardless of how many turn records it produces. We add the resulting efficiency bonus to the original task reward R_i:

$$R_i^{\text{pair}} = \begin{cases} R_i + \lambda_{\text{len}} b_i, & z_i = 1, \\ R_i, & z_i = 0. \end{cases} \tag{12}$$

Here $\lambda_{\text{len}} > 0$ controls the bonus strength, and length preferences apply only to successes. A lone success receives $b_i = 1/2$, yielding $R_i + \lambda_{\text{len}}/2$ without relative-length information. A group with no successes receives no efficiency bonus.

The comparison is invariant to a common positive rescaling of all lengths, because $\log(aT_j) - \log(aT_i) = \log T_j - \log T_i$. The score therefore depends on proportional length differences, without requiring an absolute target length. Moreover, $p_{ij} + p_{ji} = 1$, so the mean b_i over successes is $1/2$. The pairwise term allocates a bounded efficiency bonus among successful trajectories with mean bonus $\lambda_{\text{len}}/2$. Appendix A.3 proves the ordering, scale invariance, and bonus-conservation properties of this reward.

3.5 Joint Policy Optimization

We combine GRPO with selective self-distillation. The pairwise-shaped reward R_i^{pair}, adjusted for invalid actions, is normalized across turn records from the same task to obtain $A_{i,t}^{\text{RL}}$. Each response token inherits its turn's advantage. With the clipped surrogate from PPO [21], the GRPO objective is

$$\begin{aligned} \mathcal{L}_{\text{GRPO}}(\theta) = &-\mathbb{E}_{\mathcal{T}}\Big[\min\{\rho A^{\text{RL}}, \text{clip}(\rho, 1-\epsilon, 1+\epsilon)A^{\text{RL}}\}\Big] \\ &+ \beta_{\text{ref}} D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) - \lambda_H \mathcal{H}_\theta. \end{aligned} \tag{13}$$

Here $\mathbb{E}_{\mathcal{T}}$ averages over retained response tokens, with token indices omitted inside the expectation. The importance ratio is $\rho_{i,t,\ell}(\theta) = \exp(\ell^S_{i,t,\ell}(\theta) - \ell^{\text{old}}_{i,t,\ell})$, where ℓ^{old} is the sampled-token log-probability under the rollout policy. The parameter ϵ is the clipping radius. The coefficients β_{ref} and λ_H weight the KL penalty to the fixed reference policy π_{ref} and the policy entropy $\mathcal{H}_\theta$, respectively. The entropy bonus encourages continued exploration.

We add the selective OPSD loss to the RL objective:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{GRPO}}(\theta) + \lambda \mathcal{L}_{\text{OPSD}}(\theta). \tag{14}$$

The coefficient λ balances outcome-based learning with corrective supervision at selected error steps. After each outer iteration, the updated policy serves as actor and analyzer for the next round.

4 Experiments

4.1 Experimental Setup

Benchmarks and metrics. We evaluate SCOPE on ALFWorld and WebShop, following the evaluation protocols of SDAR [14]. **ALFWorld** [24] tests household task completion through text-based interaction. Each validation round contains 128 episodes from the seen split, covering six task families: Pick, Look, Clean, Heat, Cool, and Pick2. We report success rates by family and the overall success rate across the 128 episodes (micro-average). **WebShop** [36] requires agents to search, inspect, and purchase products satisfying a user's requirements. We evaluate on the same fixed set of 128 tasks and report two metrics: the mean normalized task score, which measures how well the purchased product meets the requirements, and the success rate, which counts fully completed tasks. All reported scores are percentages.

Backbones and evaluation context. We use Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct [8], and Qwen3-1.7B [33] to examine performance across model sizes and generations. Comparisons are made within each backbone. At evaluation, SCOPE acts from the ordinary interaction history without hindsight skills or analyzer calls. For the main ALFWorld and WebShop comparison, we report SCOPE's validation metrics logged at training step 150. An asterisk marks a baseline that receives skills at evaluation; the other baselines use their ordinary policy inputs. Key training settings and diagnostic protocols are provided in Appendix B.

Baselines. We compare against eight baselines. **Vanilla** evaluates the original backbone without additional agentic training. **Skill-Prompt*** supplies retrieved skills in the evaluation prompt without updating the model, following the prompting baseline in SDAR [14]. **GRPO** [22] learns from group-relative task rewards without an auxiliary distillation objective.

The remaining methods incorporate self-distillation or hindsight guidance into agentic RL. **AgentOPSD** [28] aggregates teacher–student gaps by turn and uses recursive belief updates to assign turn-level credit. **SDAR** [14] retrieves teacher guidance from a skill bank and controls auxiliary self-distillation with bounded token-level gates. **OPID** [35] uses an external analyzer to construct hierarchical skills, retaining episode-level guidance at non-critical steps. **GRSD** [43] uses policy-generated reflections scored by an external judge to construct group-level guidance and refine turn-level credit assignment. **SEED** [29] initializes hindsight analysis through SFT and iteratively uses the updated policy to generate and distill trajectory-level skills.

4.2 Main Results

Table 1 compares SCOPE with eight baselines across three backbones, grouped by their auxiliary guidance types. SCOPE achieves the highest WebShop success across all backbones and the highest or tied highest ALFWorld success.

Beyond outcome-level supervision. Compared with GRPO, SCOPE improves ALFWorld success by 22.7, 17.2, and 46.1 percentage points on Qwen2.5-3B, Qwen2.5-7B, and Qwen3-1.7B, respectively. The corresponding WebShop gains are 21.9, 11.8, and 40.6 points. These consistent improvements suggest the benefit of complementing coarse outcome rewards with localized corrective supervision, rather than relying solely on trajectory-level feedback.

Self-bootstrapped error analysis versus external supervision. OPID relies on an external analyzer to identify critical steps and generate hierarchical skills, while GRSD uses an external judge to evaluate policy-generated reflections. In contrast, SCOPE internalizes error localization and skill generation through supervised initialization, enabling the policy to provide its own corrective guidance during online RL. Compared with OPID, SCOPE improves ALFWorld success by 13.4, 8.4, and 33.3 percentage points across the three backbones. It also outperforms GRSD by 11.0 and 6.2 points on Qwen2.5-3B and Qwen2.5-7B, respectively, while matching its ALFWorld success on Qwen3-1.7B. These results support the effectiveness of self-bootstrapped corrective guidance without external online analysis.

Selective step guidance versus trajectory-level skills. Compared with trajectory-level SEED, SCOPE improves ALFWorld success by 5.9 and 2.3 points on the Qwen2.5-3B and Qwen2.5-7B backbones, respectively, while gaining 6.3 points on WebShop for both. On Qwen3-1.7B, SCOPE ties GRSD at 92.2% ALFWorld success and exceeds SEED and GRSD on WebShop success. Notably, SCOPE achieves the highest WebShop success across all backbones without consistently leading in normalized task score, suggesting stronger exact task completion rather than uniformly better partial matches. These results are consistent with the value of aligning distillation with selected error steps instead of applying trajectory-level guidance broadly.

4.3 Diagnosing Trajectory-Level Supervision

Figure 1(a) reveals substantial disagreement between trajectory outcomes and step quality in GLM-5.3 [40]-annotated WebShop trajectories. Moreover, Figure 1(b) shows that skill-unrelated steps exhibit teacher–student gaps comparable in magnitude to those at relevant steps ($-0.378/+0.230$ versus $-0.400/+0.249$ nats). These findings motivate error localization and selective distillation rather than relying on outcome labels or gap magnitude alone. Detailed protocols are provided in Appendix B.3.

4.4 Ablation Studies

We evaluate six variants with Qwen2.5-3B on ALFWorld (Table 2) to examine the contributions of SCOPE's components and different forms of guidance. Together with the parameter sweeps in Section 4.6, these experiments also examine the effect of guidance strength. All endpoint comparisons in this section use training step 180, where the full configuration achieves 98.4% overall success.

Supervised initialization supports subsequent self-analysis. Removing SFT initialization lowers overall success from 98.4% to 92.2%, a 6.3-point drop. Its effect is also visible during training: at step 100, the full configuration reaches 96.1%, compared with 78.1% without SFT (Figure 4(A)). Heat is especially affected at the final checkpoint, falling from 100.0% to 70.6%. These results highlight the need to first equip the model to locate errors and generate corrective guidance before relying on its own analyses for online supervision.

Matching skills to target steps improves guidance. Replacing step-specific skills and selective coverage with one skill per trajectory and distillation over all steps reduces success to 96.1%, a 2.3-point drop. Replacing the generated guidance with skill-bank guidance yields 95.3%, a 3.1-point drop. Together with the unrelated-step signals in Section 4.3, these results support constructing guidance from the current trajectory and restricting its application to the relevant decisions. The trajectory-level variant changes skill granularity and coverage jointly, so its result supports their combined design.

Success-relative pairwise rewards adapt to policy improvement. Removing the length reward reduces overall success from 98.4% to 93.0%, while replacing the pairwise reward with an absolute length bonus yields 97.7%. Unlike absolute length shaping, which depends on fixed length scales, our pairwise reward compares successful trajectories within each on-policy group, adapting its efficiency preferences as the policy improves. Its fixed mean bonus redistributes rewards toward shorter successes without imposing an absolute length target. Figure 4(A) shows that the pairwise design maintains competitive success throughout training and achieves the highest final performance.

4.5 Training Dynamics

SCOPE reaches strong performance earlier in online training. The WebShop curves in Figure 1(c) show that SCOPE reaches 65.1% mean success at step 60, exceeding GRPO's 63.0% at its final exported checkpoint, step 150. The first observed crossing occurs 90 updates earlier, a 60.0% reduction. On ALFWorld, the SCOPE run shown in Figure 1(c), left, first exceeds the final GRPO mean success rate of 76.6% at step 20, compared with step 145 for GRPO, using 86.2% fewer online updates. At step 180, this SCOPE run reaches 98.4% versus 76.6% for GRPO, a gain of 21.9 percentage points.

Table 1: Main results on ALFWorld and WebShop (%). ALFWorld reports task success; WebShop reports normalized task score and success rate. An asterisk denotes skills at evaluation. Type describes the auxiliary guidance. Hybrid skills combine trajectory- and step-level skills; Traj. denotes trajectory-level. Best and second-best available results are highlighted.

		ALFWorld							WebShop	
Method	**Type**	**Pick**	**Look**	**Clean**	**Heat**	**Cool**	**Pick2**	**Avg**	**Score**	**Succ.**
Qwen2.5-3B-Instruct										
Vanilla (? , ?)	None	44.4	11.1	6.2	15.4	28.6	12.5	21.9	6.7	0.8
Skill-Prompt* (? , ?)	Skill bank	51.7	66.7	48.4	0.0	4.3	10.0	28.9	0.2	0.8
GRPO (Shao et al., 2024)	None	91.2	62.5	96.2	61.9	65.0	47.4	75.0	79.8	63.3
AgentOPSD (Wang et al., 2026a)	Skill bank	**100.0**	68.8	86.4	66.7	84.6	73.7	84.4	**90.4**	69.5
SDAR (? , ?)	Skill bank	97.1	62.5	**100.0**	61.9	75.0	84.2	84.4	85.0	68.0
OPID (Yang et al., 2026b)	Hybrid skills	92.7	**100.0**	88.9	70.0	84.2	70.0	84.3	85.0	74.2
GRSD (? , ?)	Traj. skill	**100.0**	81.3	91.3	88.2	70.8	85.7	86.7	86.9	76.5
SEED (Wu et al., 2026a)	Traj. skill	**100.0**	**100.0**	**100.0**	**100.0**	70.6	80.0	91.8	88.5	78.9
SCOPE (Ours)	**Step skills**	**100.0**	**100.0**	**100.0**	**100.0**	**85.7**	**100.0**	**97.7**	87.9	**85.2**
Qwen2.5-7B-Instruct										
Vanilla (? , ?)	None	36.1	22.2	3.1	0.0	0.0	0.0	12.5	5.9	1.6
Skill-Prompt* (? , ?)	Skill bank	51.7	50.0	32.3	5.3	4.3	0.0	23.4	1.7	0.8
GRPO (Shao et al., 2024)	None	91.2	87.5	96.2	81.0	65.0	57.9	81.2	80.9	72.6
AgentOPSD (Wang et al., 2026a)	Skill bank	91.2	87.5	**100.0**	90.5	75.0	84.2	89.1	90.2	79.7
SDAR (? , ?)	Skill bank	94.7	75.0	**100.0**	86.7	68.2	78.9	85.9	89.4	82.8
OPID (Yang et al., 2026b)	Hybrid skills	**100.0**	81.8	97.1	**100.0**	80.8	80.0	90.0	85.3	79.7
GRSD (? , ?)	Traj. skill	**100.0**	**100.0**	**100.0**	93.4	66.7	82.4	92.2	**91.6**	82.8
SEED (Wu et al., 2026a)	Traj. skill	**100.0**	**100.0**	96.3	80.0	**100.0**	**100.0**	96.1	89.7	78.1
SCOPE (Ours)	**Step skills**	**100.0**	**100.0**	**100.0**	**100.0**	95.2	94.7	**98.4**	89.6	**84.4**
Qwen3-1.7B										
Vanilla (Yang et al., 2025)	None	25.0	22.2	3.1	0.0	21.4	4.2	12.5	46.5	4.7
Skill-Prompt* (? , ?)	Skill bank	10.3	50.0	16.1	0.0	0.0	5.0	9.4	23.0	2.3
GRPO (Shao et al., 2024)	None	71.1	41.7	36.4	40.0	31.8	31.6	46.1	67.3	38.3
AgentOPSD (Wang et al., 2026a)	Skill bank	35.5	50.0	31.6	30.0	46.2	25.0	35.9	66.8	47.7
SDAR (? , ?)	Skill bank	73.5	25.0	76.9	33.3	40.0	36.8	53.9	76.8	58.6
OPID (Yang et al., 2026b)	Hybrid skills	65.9	72.7	66.7	40.0	63.2	45.0	58.9	79.6	64.8
GRSD (? , ?)	Traj. skill	92.5	**100.0**	**100.0**	91.3	81.3	88.2	**92.2**	84.5	66.4
SEED (Wu et al., 2026a)	Traj. skill	97.6	**100.0**	**100.0**	80.0	**84.2**	**90.0**	92.0	**87.1**	77.3
SCOPE (Ours)	**Step skills**	**100.0**	**100.0**	96.9	**100.0**	66.7	89.5	**92.2**	83.2	**78.9**

These comparisons measure online policy updates after supervised initialization.

Auxiliary distillation requires controlled optimization. Figure 3 shows an extreme training-stage gap alongside a sharp drop in current student log-probabilities during minibatch optimization, while cached teacher scores remain moderate. The full SCOPE run in Figure 4 maintains high success late in training. The coefficient bound in Equation 8 limits gap-driven amplification; Section 4.6 further examines its effect on task performance.

4.6 Parameter Sensitivity

We vary the gap slope α, saturation bound c, and pairwise temperature τ_{len} individually, keeping the other parameters at $\alpha = 2.5$, $c = 1$, and $\tau_{\text{len}} = 0.1$. Figure 5 breaks down ALFWorld validation success by task family at training step 180.

The signed-gap slope controls useful update strength. At $\alpha = 1$, overall success is 89.8%, compared with 98.4% at the default $\alpha = 2.5$ and 97.7% at $\alpha = 5$. The smaller slope reduces the coefficient near zero in Equation 8; its weaker endpoint performance is consistent with insufficient auxiliary guidance in this setting. The task breakdown shows the largest losses on Cool and Pick2. Increasing the slope to 5 remains competitive, suggesting that the benefit is not confined to the exact default value.

Overly tight saturation limits correction. Reducing c to 0.1 lowers success to 94.5%, with Heat at 82.4%. Both $c = 1$ and $c = 10$ reach 98.4% overall success, indicating a range of effective saturation bounds. All three configurations use the bounded signed tanh transformation.

A moderate pairwise temperature performs best in the sweep. Temperatures of 0.05 and 0.2 achieve 93.8% and 96.1%, respectively,

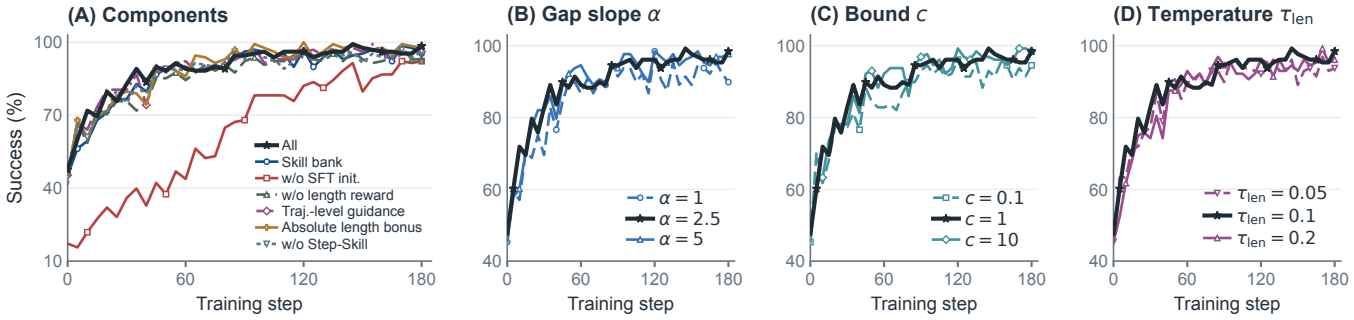


Figure 4: Training dynamics on ALFWorld with Qwen2.5-3B. (A) Component ablations. (B) Signed-gap slope α. (C) Tanh saturation bound c. (D) Pairwise length temperature τ_{len}. The full setting is $\alpha = 2.5$, $c = 1$, and $\tau_{\text{len}} = 0.1$, labeled All in (A) and by its parameter value in (B)–(D). Trajectory-level guidance uses one skill per trajectory and distills all steps. Lines connect observed checkpoints; the success axes start at 10% in (A) and 40% in (B)–(D).

Table 2: Ablation results on ALFWorld (%). Qwen2.5-3B at step 180. Overall is the success rate across validation episodes. Best values, including ties, are bold.

	ALFWorld						
Variant	**Pick**	**Look**	**Clean**	**Heat**	**Cool**	**Pick2**	**Overall**
Full SCOPE	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	92.0	**98.4**
Initialization and guidance							
w/o SFT initialization	**100.0**	92.9	**100.0**	70.6	83.3	**94.7**	92.2
w/o Step-Skill	**100.0**	92.9	**100.0**	94.1	94.4	89.5	96.1
Traj.-level guidance	**100.0**	92.9	**100.0**	88.2	**100.0**	89.5	96.1
Skill-bank guidance	**100.0**	**100.0**	96.6	76.5	**100.0**	**94.7**	95.3
Length-reward design							
w/o length reward	96.8	91.7	**100.0**	**100.0**	92.3	75.0	93.0
Absolute length bonus	**100.0**	**100.0**	**100.0**	94.1	**100.0**	89.5	97.7

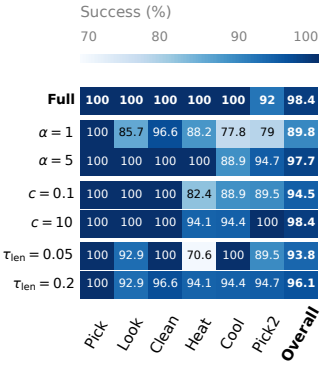


Figure 5: Parameter sensitivity on ALFWorld (%, step 180).

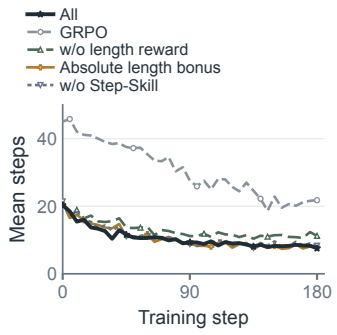


Figure 6: Mean response steps on ALFWorld. GRPO: seed0. Lower is better.

compared with 98.4% at $\tau_{\text{len}} = 0.1$. A lower temperature makes pairwise length preferences more abrupt, whereas a higher temperature softens their differences. The intermediate setting yields the highest observed overall success, supporting moderate length discrimination alongside the corrective objective.

4.7 Generalization to Unseen Tasks

The gains extend to unseen task instances. Figure 7 reports 89.6% aggregate success for SCOPE versus 66.4% for GRPO, a gain of 23.2 percentage points. SCOPE improves all six task families, with particularly large gains on Look (+73.3 points), Pick2 (+50.0), and Cool (+32.4). The improvement across different household routines supports the transfer of learned behavior beyond the seen evaluation tasks.

Higher completion coincides with fewer interactions. The reported average number of turns falls from 18.86 to 10.06, a 46.7% reduction, alongside the increase in success. This joint pattern supports the efficiency motivation in Section 1: the complete SCOPE policy achieves more tasks with shorter reported interactions. The length-reward ablation complements this observation by showing that the pairwise preference also contributes to completion performance.

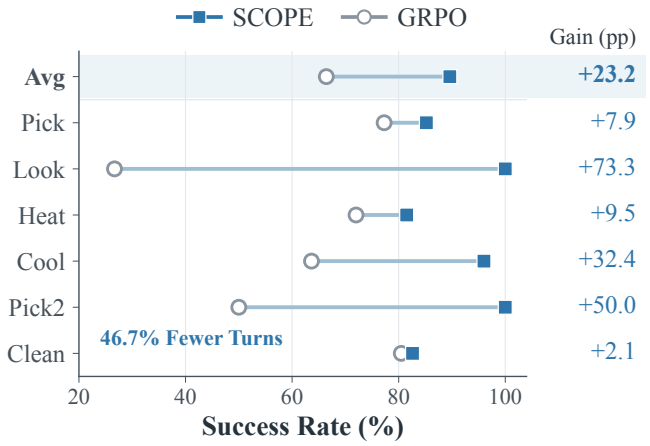


Figure 7: Generalization on ALFWorld Unseen. Blue squares denote SCOPE and gray circles denote GRPO. Gains are absolute success-rate differences in percentage points. The in-plot statement summarizes the relative reduction in reported average turns, from 18.86 to 10.06 (46.7%).

5 Conclusion

SCOPE aligns corrective self-distillation with the decisions that require it. Error localization separates intermediate decision quality from trajectory outcomes, while step-specific skills restrict privileged guidance to relevant steps. Bounded signed coefficients control distillation strength, and success-relative length rewards encourage efficient completion. Experiments on ALFWorld and WebShop show improvements over outcome-only RL across three backbones, with further gains on unseen tasks and reduced interaction counts. The resulting policy performs tasks without skill conditioning or analyzer calls at inference time.

References

[1] Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos Garea, Matthieu Geist, and Olivier Bachem. 2024. On-policy distillation of language models: Learning from self-generated mistakes. In *International Conference on Learning Representations*, Vol. 2024. 21246–21263.

[2] Pranjal Aggarwal and Sean Welleck. 2025. L1: Controlling how long a reasoning model thinks with reinforcement learning. *arXiv preprint arXiv:2503.04697* (2025).

[3] Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, OpenAI Pieter Abbeel, and Wojciech Zaremba. 2017. Hindsight experience replay. *Advances in neural information processing systems* 30 (2017).

[4] Ralph Allan Bradley and Milton E Terry. 1952. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika* 39, 3/4 (1952), 324–345.

[5] Xin Cheng, Shuo He, Lang Feng, HaiYang Xu, Ming Yan, Lei Feng, and Bo An. 2026. Beyond Trajectory-Level Attribution: Graph-Based Credit Assignment for Agentic Reinforcement Learning. *arXiv preprint arXiv:2605.26684* (2026).

[6] Lang Feng, Zhenghai Xue, Tingcong Liu, and Bo An. 2026. Group-in-group policy optimization for llm agent training. *Advances in Neural Information Processing Systems* 38 (2026), 46375–46408.

[7] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531* (2015).

[8] Binyuan Hui, Jian Yang, Zeyu Cui, Jiaxi Yang, Dayiheng Liu, Lei Zhang, Tianyu Liu, Jiajun Zhang, Bowen Yu, Keming Lu, et al. 2024. Qwen2. 5-coder technical report. *arXiv preprint arXiv:2409.12186* (2024).

[9] Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Sercan Arik, Dong Wang, Hamed Zamani, and Jiawei Han. 2025. Search-r1: Training llms to reason and leverage search engines with reinforcement learning. *arXiv preprint arXiv:2503.09516* (2025).

[10] Yoon Kim and Alexander M Rush. 2016. Sequence-level knowledge distillation. In *Proceedings of the 2016 conference on empirical methods in natural language processing*. 1317–1327.

[11] Qirui Liu, Yichen Sun, Yan Wang, Zhixuan Chu, Linbo Jiang, Jianan Lin, and Kui Ren. 2026. DeShortcut-Align: Decoupling Spurious Shortcuts for Robust Safety Alignment in Large Reasoning Models. *arXiv preprint arXiv:2609.34896* (2026).

[12] Qirui Liu, Yichen Sun, Yan Wang, Yu Mi, Wei Cao, Yue Shen, Zhixuan Chu, and Kui Ren. 2026. EOPSA: Efficient On-Policy Self-Distilled Safety Alignment. *arXiv preprint arXiv:2609.34519* (2026).

[13] Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, et al. 2024. Agentbench: Evaluating llms as agents. In *International Conference on Learning Representations*, Vol. 2024. 52989–53046.

[14] Zhengxi Lu, Zhiyuan Yao, Zhuowen Han, Zi-Han Wang, Jinyang Wu, Qi Gu, Xunliang Cai, Weiming Lu, Jun Xiao, Yueting Zhuang, et al. 2026. Self-distilled agentic reinforcement learning. *arXiv preprint arXiv:2605.15155* (2026).

[15] Junyu Luo, Weizhi Zhang, Ye Yuan, Yusheng Zhao, Junwei Yang, Yiyang Gu, Bohan Wu, Binqi Chen, Ziyue Qiao, Qingqing Long, et al. 2025. Large language model agent: A survey on methodology, applications and challenges. *arXiv preprint arXiv:2503.21460* (2025).

[16] Xufang Luo, Yuge Zhang, Zhiyuan He, Zilong Wang, Siyun Zhao, Dongsheng Li, Luna K Qiu, and Yuqing Yang. 2025. Agent lightning: Train any ai agents with reinforcement learning. *arXiv preprint arXiv:2508.03680* (2025).

[17] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. 2023. Self-refine: Iterative refinement with self-feedback. *Advances in neural information processing systems* 36 (2023), 46534–46594.

[18] Cheng Qian, Emre Can Acikgoz, Qi He, Hongru Wang, Xiusi Chen, Dilek Hakkani-Tur, Gokhan Tur, and Heng Ji. 2026. Toolrl: Reward is all tool learning needs. *Advances in Neural Information Processing Systems* 38 (2026), 105523–105553.

[19] Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. 2011. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the fourteenth international conference on artificial intelligence and statistics*. JMLR Workshop and Conference Proceedings, 627–635.

[20] Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. *Advances in neural information processing systems* 36 (2023), 68539–68551.

[21] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347* (2017).

[22] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300* (2024).

[23] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. *Advances in neural information processing systems* 36 (2023), 8634–8652.

[24] Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. 2020. Alfworld: Aligning text and embodied environments for interactive learning. *arXiv preprint arXiv:2010.03768* (2020).

[25] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2023. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291* (2023).

[26] Hao Wang, Guozhi Wang, Han Xiao, Yufeng Zhou, Yue Pan, Jichao Wang, Ke Xu, Yafei Wen, Xiaohu Ruan, Xiaoxin Chen, et al. 2026. Skill-sd: Skill-conditioned self-distillation for multi-turn llm agents. *arXiv preprint arXiv:2604.10674* (2026).

[27] Zihan Wang, Kangrui Wang, Qineng Wang, Pingyue Zhang, Linjie Li, Zhengyuan Yang, Xing Jin, Kefan Yu, Minh Nhat Nguyen, Licheng Liu, et al. 2025. Ragen: Understanding self-evolution in llm agents via multi-turn reinforcement learning. *arXiv preprint arXiv:2504.20073* (2025).

[28] Zi-Han Wang, Zhengxi Lu, Zhiyuan Yao, Jinyang Wu, Jie Wu, Zhengzhou Cai, Yueqing Sun, Ziang Ye, Linji Hao, Qi Gu, et al. 2026. Agentopsd: Recursive self-distillation for agentic reinforcement learning. *arXiv preprint arXiv:2608.05987* (2026).

[29] Jinyang Wu, Shuo Yang, Zhengxi Lu, Fan Zhang, Yuhao Shen, Lang Feng, Haoran Luo, Zheng Lian, Shuai Zhang, Zhengqi Wen, et al. 2026. SEED: Self-Evolving On-Policy Distillation for Agentic Reinforcement Learning. *arXiv preprint arXiv:2607.14777* (2026).

[30] Jinyang Wu, Shuo Yang, Yuhao Shen, Shuai Zhang, Zhengqi Wen, and Jianhua Tao. 2026. Spark: Strategic policy-aware exploration via dynamic branching for long-horizon agentic learning. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 23981–24004.

[31] Jinyang Wu, Guocheng Zhai, Ruihan Jin, Jiahao Yuan, Yuhao Shen, Shuai Zhang, Zhengqi Wen, and Jianhua Tao. 2026. Atlas: Orchestrating heterogeneous models and tools for multi-domain complex reasoning. In *Findings of the Association for Computational Linguistics: ACL 2026*. 17503–17535.

[32] Hang Yan, Fangzhi Xu, Qiushi Sun, Jinyang Wu, Zixian Huang, Muye Huang, Jingyang Gong, Zichen Ding, Kanzhi Cheng, Yian Wang, et al. 2026. Odysseyarena: Benchmarking large language models for long-horizon, active and inductive interactions. *arXiv preprint arXiv:2602.05843* (2026).

[33] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388* (2025).

[34] Chenxu Yang, Chuanyu Qin, Qingyi Si, Minghui Chen, Naibin Gu, Dingyu Yao, Zheng Lin, Weiping Wang, Jiaqi Wang, and Nan Duan. 2026. Self-distilled rlvr. *arXiv preprint arXiv:2604.03128* (2026).

[35] Shuo Yang, Jinyang Wu, Zhengxi Lu, Yuhao Shen, Fan Zhang, Lang Feng, Shuai Zhang, Haoran Luo, Zheng Lian, Zhengqi Wen, et al. 2026. Opid: On-policy skill distillation for agentic reinforcement learning. *arXiv preprint arXiv:2606.26790* (2026).

[36] Shunyu Yao, Howard Chen, John Yang, and Karthik Narasimhan. 2022. Webshop: Towards scalable real-world web interaction with grounded language agents. *Advances in Neural Information Processing Systems* 35 (2022), 20744–20757.

[37] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2022. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629* (2022).

[38] Jingyang Yi, Jiazheng Wang, and Sida Li. 2026. Shorterbetter: Guiding reasoning models to find optimal inference length for efficient reasoning. *Advances in Neural Information Processing Systems* 38 (2026), 39011–39043.

[39] Z.ai. 2026. GLM-5.2: Built for Long-Horizon Tasks. Accessed: 2026-10-23. https://z.ai/blog/glm-5.2

[40] Z.ai. 2026. GLM-5.3: Frontier Coding with Emergent Cyber Capabilities. Accessed: 2026-10-23. https://z.ai/blog/glm-5.3

[41] Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, and Gao Huang. 2024. Expel: Llm agents are experiential learners. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 19632–19642.

[42] Siyan Zhao, Zhihui Xie, Mengchen Liu, Jing Huang, Guan Pang, Feiyu Chen, and Aditya Grover. 2026. Self-distilled reasoner: On-policy self-distillation for large language models. *arXiv preprint arXiv:2601.18734* (2026).

[43] Binbin Zheng, Zijun Xie, Guanqun Zhao, Enlei Gong, Xing Ma, Xiaoliang Fu, and Zeyu Chen. 2026. Group-reflective self-distillation for agentic reinforcement learning. *arXiv preprint arXiv:2607.28076* (2026).

[44] Qiyong Zhong, Mao Zheng, Mingyang Song, Xin Lin, Jie Sun, Houcheng Jiang, Xiang Wang, and Junfeng Fang. 2026. Sod: Step-wise on-policy distillation for small language model agents. *arXiv preprint arXiv:2605.07725* (2026).

A Theoretical Analysis

SCOPE addresses three design questions: how to preserve signed correction while controlling extreme teacher–student discrepancies, where to apply that correction, and how to reward efficient task completion. We analyze these choices in turn and relate each result to the diagnostics, ablations, or parameter sweeps in Section 4.5, Section 4.4, and Section 4.6.

Setting and notation. For the distillation analysis, fix one training microbatch, its sampled responses and prefixes, the assigned skills, the retained-token mask, and the cached teacher log-probabilities. Let $\mathcal{I} = \mathcal{T}_{\text{sel}}$ denote the selected token set, with $n = |\mathcal{I}| > 0$. Each $k \in \mathcal{I}$ abbreviates a token index (i, t, ℓ). Write

$$\begin{aligned} r_k(\theta) &= \ell_k^T - \ell_k^S(\theta), \qquad v_k(\theta) = \nabla_\theta \ell_k^S(\theta), \\ f(r) &= c \tanh(\alpha r/c), \qquad \alpha, c > 0. \end{aligned} \tag{15}$$

Here r_k denotes the differentiable log-probability difference; Equation 8 uses its detached value to form the coefficient. All log-probabilities are finite, and student log-probabilities are differentiable at the parameters under study. The gradient produced by Equation 9 is therefore

$$g(\theta) = -\frac{1}{n} \sum_{k \in \mathcal{I}} f(r_k(\theta)) v_k(\theta). \tag{16}$$

If $n = 0$, the auxiliary loss and gradient are defined to be zero. All derivatives below use this fixed-data setting, including fixed selection and filtering decisions.

A.1 Bounded Signed Distillation

Why use a saturating signed coefficient? The coefficient should preserve the direction of corrective guidance and remain responsive near agreement, while limiting the influence of extreme discrepancies. Its gradient-equivalent potential makes this tradeoff explicit.

Proposition A.1 (Gradient equivalence to a log-cosh potential). *Under the fixed-microbatch setting above, define*

$$\begin{aligned} \varphi(r) &= \frac{c^2}{\alpha} \log\cosh(\alpha r/c), \\ \Phi(\theta) &= \frac{1}{n} \sum_{k \in \mathcal{I}} \varphi(r_k(\theta)). \end{aligned} \tag{17}$$

Then $\nabla_\theta \Phi(\theta) = g(\theta)$. Moreover, φ is nonnegative, even, and strictly convex in its scalar argument, with its unique minimum at $r = 0$. Its limiting forms are

$$\begin{aligned} \varphi(r) &= \frac{\alpha}{2} r^2 + O(r^4), && r \to 0, \\ \varphi(r) &= c|r| - \frac{c^2}{\alpha} \log 2 + o(1), && |r| \to \infty. \end{aligned} \tag{18}$$

Proof. Differentiating with respect to r gives

$$\varphi'(r) = f(r), \qquad \varphi''(r) = \frac{\alpha}{\cosh^2(\alpha r/c)} > 0. \tag{19}$$

Since the cached teacher is fixed, $\nabla_\theta r_k = -v_k$. Applying the chain rule yields

$$\nabla_\theta \Phi = \frac{1}{n} \sum_{k \in \mathcal{I}} \varphi'(r_k)(-v_k) = g. \tag{20}$$

The inequality $\cosh x \geq 1$, with equality only at $x = 0$, and the evenness of cosh establish nonnegativity, symmetry, and the unique scalar minimum. Finally, $\log\cosh x = x^2/2 + O(x^4)$ near zero, while

$$\log\cosh x = |x| - \log 2 + \log(1 + e^{-2|x|}).$$

Substitution proves Equation 18. □

Proposition A.1 gives the detached update a log-cosh interpretation on the current microbatch. The quadratic region retains a response proportional to small discrepancies, while the linear tails limit their influence as the gap grows. Oddness of f preserves opposite coefficients for reinforcement and suppression. The slope α controls the local response; the sweep in Figure 5 shows 89.8% success at $\alpha = 1$, 98.4% at 2.5, and 97.7% at 5, connecting this local response to the observed sensitivity of task completion.

How is gap-driven amplification controlled? The instability diagnostic in Figure 3 shows that a cached teacher can coexist with extreme training-stage gaps as the student changes. The next result quantifies how saturation limits their contribution to the auxiliary gradient.

Proposition A.2 (Bounded coefficients and gradients). *For every finite r, the coefficient satisfies*

$$\begin{aligned} &|f(r)| < c, \qquad f(0) = 0, \\ &\operatorname{sign} f(r) = \operatorname{sign} r, \\ &0 < f'(r) \leq \alpha. \end{aligned} \tag{21}$$

If $\|v_k\| \leq B_k$ for each selected token, then

$$\|g\| \leq \frac{c}{n} \sum_{k \in \mathcal{I}} B_k. \tag{22}$$

In particular, $\|g\| \leq cB$ when $B_k \leq B$ uniformly.

Proof. The range, oddness, and strict monotonicity of tanh imply the first three properties. Differentiating gives $f'(r) = \alpha/\cosh^2(\alpha r/c) \in (0, \alpha]$. Applying the triangle inequality to Equation 16 gives

$$\|g\| \leq \frac{1}{n} \sum_{k \in \mathcal{I}} |f(r_k)| \|v_k\| \leq \frac{c}{n} \sum_{k \in \mathcal{I}} B_k.$$

□

Under the uniform score-gradient bound, the auxiliary contribution to the joint objective is bounded by $\lambda_{\text{OPSD}} cB$. Thus c directly controls the maximum amplification attributable to the gap. The saturation sweep in Figure 5 complements this bound: tightening c from 1 to 0.1 lowers success from 98.4% to 94.5%, while $c = 10$ also reaches 98.4%. Together, the bound and sweep explain the role of c: limiting extreme coefficients while retaining sufficient corrective strength.

How sensitive is the update to teacher-score perturbations? To isolate the effect of the cached supervision, consider changing teacher scores while keeping the sampled data, student, and selected positions fixed.

Corollary A.3 (Sensitivity to cached teacher scores). *Keep the student parameters, sampled tokens, and selected set fixed. Suppose*

the cached teacher scores change from ℓ_k^T to $\ell_k^T + e_k$, and let $\widetilde{g}$ be the resulting auxiliary gradient. Under $\|v_k\| \le B$,

$$\|\widetilde{g} - g\| \le \frac{B}{n} \sum_{k \in \mathcal{I}} \min\{\alpha|e_k|, 2c\}. \tag{23}$$

Proof. The derivative bound in Proposition A.2 and the mean value theorem give $|f(r+e) - f(r)| \le \alpha|e|$. Boundedness also gives $|f(r+e) - f(r)| \le 2c$. Apply both bounds to

$$\widetilde{g} - g = -\frac{1}{n} \sum_{k \in \mathcal{I}} [f(r_k + e_k) - f(r_k)] v_k$$

and use the triangle inequality. □

The perturbation bound separates the roles of the two parameters: α bounds local sensitivity, and $2c$ caps the coefficient change caused by an arbitrarily large finite score perturbation. This supplies an update-level interpretation of the separate α and c sweeps in Figure 5; those experiments evaluate the downstream task performance of the resulting configurations.

A.2 Influence of Irrelevant Selected Tokens

Why restrict distillation to selected steps? Section 4.3 identifies skill-conditioned signals at decisions unrelated to the skill. We quantify the direct contribution of such signals by partitioning the retained selected tokens according to skill relevance. Let $\mathcal{I}_{\text{off}} \subseteq \mathcal{I}$ contain tokens at steps for which the assigned skill is irrelevant, and let $\mathcal{I}_{\text{rel}} = \mathcal{I} \setminus \mathcal{I}_{\text{off}}$. For this analytical partition, define $\eta_{\text{off}} = |\mathcal{I}_{\text{off}}|/n$. The fraction is token-weighted, matching the loss normalization in Equation 9.

Proposition A.4 (Off-target gradient contribution). *Under $\|v_k\| \le B$, the contribution of irrelevant selected tokens,*

$$g_{\text{off}} = -\frac{1}{n} \sum_{k \in \mathcal{I}_{\text{off}}} f(r_k) v_k, \tag{24}$$

satisfies

$$\|g_{\text{off}}\| \le cB\eta_{\text{off}}. \tag{25}$$

If $\mathcal{I}_{\text{rel}}$ is nonempty, define the mean gradient over the relevant selected tokens by

$$\bar{g}_{\text{rel}} = -\frac{1}{|\mathcal{I}_{\text{rel}}|} \sum_{k \in \mathcal{I}_{\text{rel}}} f(r_k) v_k. \tag{26}$$

Then the effect of including the irrelevant tokens, accounting for the change in normalization, obeys

$$\|g - \bar{g}_{\text{rel}}\| \le 2cB\eta_{\text{off}}. \tag{27}$$

Proof. Each summand has norm at most cB. Summing over $|\mathcal{I}_{\text{off}}|$ tokens and dividing by n proves Equation 25. The full gradient decomposes as

$$g = (1 - \eta_{\text{off}})\bar{g}_{\text{rel}} + g_{\text{off}}.$$

Since $\|\bar{g}_{\text{rel}}\| \le cB$, subtraction and the triangle inequality give

$$\|g - \bar{g}_{\text{rel}}\| \le \eta_{\text{off}}\|\bar{g}_{\text{rel}}\| + \|g_{\text{off}}\| \le 2cB\eta_{\text{off}}.$$

□

Proposition A.4 connects the two controls in SCOPE: step selection determines which tokens enter the loss, while saturation bounds each token's contribution. A smaller irrelevant-token fraction tightens the worst-case deviation from the mean update over the relevant selected tokens. The comparison retains the same relevant tokens and scores, isolating the effect of including irrelevant ones. Shared parameters still couple predictions across steps.

The trajectory-level ablation in Table 2 provides the corresponding performance comparison: replacing step-specific skills and selective coverage with a single skill per trajectory and all-step distillation reduces success from 98.4% to 96.1%. This experiment evaluates skill granularity and coverage jointly. The unrelated-step diagnostic motivates restricting coverage, the proposition describes how irrelevant selected tokens can affect the update, and the ablation evaluates the combined design in task completion.

A.3 Properties of Success-Relative Efficiency Rewards

How can efficiency be rewarded without an absolute length target? SCOPE compares successful trajectories from the same task. The reward should favor shorter successes, depend on relative lengths, and allocate a controlled bonus within that group. Fix a rollout group and its success set $\mathcal{S}_q$, and let $M = |\mathcal{S}_q|$. Assume all successful interaction lengths are finite and positive, and $\tau_{\text{len}} > 0$. The quantities p_{ij}, b_i, and R_i^{pair} follow Equations 10–12.

Proposition A.5 (Ordering, scale invariance, and bonus conservation). *For $M \ge 2$, the scores satisfy the following properties:*

(i) $0 < b_i < 1$, and, for distinct successful trajectories, $T_i < T_j$ implies $b_i > b_j$, while $T_i = T_j$ implies $b_i = b_j$.
(ii) Multiplying every successful length by the same $a > 0$ leaves all pairwise probabilities and scores unchanged.
(iii) The total score and efficiency bonus are

$$\sum_{i \in \mathcal{S}_q} b_i = \frac{M}{2}, \qquad \sum_{i \in \mathcal{S}_q} (R_i^{\text{pair}} - R_i) = \frac{\lambda_{\text{len}} M}{2}. \tag{28}$$

The identities in (iii) also hold for $M = 1$ with $b_i = 1/2$, and for $M = 0$ with empty sums.

Proof. For finite positive lengths, each sigmoid value lies strictly between zero and one, and so does its average b_i. To compare two scores, expand their difference:

$$(M-1)(b_i - b_j) = p_{ij} - p_{ji} + \sum_{k \in \mathcal{S}_q \setminus \{i,j\}} (p_{ik} - p_{jk}). \tag{29}$$

If $T_i < T_j$, then $p_{ij} > 1/2 > p_{ji}$. For every common opponent k, strict monotonicity of the sigmoid gives $p_{ik} > p_{jk}$. The right-hand side is therefore positive, including when $M = 2$ and the sum is empty. If the two lengths are equal, all differences vanish. This proves (i).

For (ii), replacing T_i, T_j by aT_i, aT_j leaves $\log T_j - \log T_i$ unchanged. Thus every p_{ij}, and hence every b_i, is unchanged.

For (iii), the identity $\sigma(x) + \sigma(-x) = 1$ gives $p_{ij} + p_{ji} = 1$. Grouping ordered comparisons into unordered pairs,

$$\sum_{i \in \mathcal{S}_q} b_i = \frac{1}{M-1} \sum_{\substack{i,j \in \mathcal{S}_q \\ i<j}} (p_{ij} + p_{ji}) = \frac{\binom{M}{2}}{M-1} = \frac{M}{2}. \tag{30}$$

Multiplication by λ_{len} proves the bonus identity. The definitions for a single success and an empty success set give the remaining cases directly. □

The proposition establishes a relative efficiency preference: shorter successes receive higher scores, and common rescaling of interaction lengths leaves the scores unchanged. For each success, the reward can be decomposed as

$$R_i^{\text{pair}} = R_i + \frac{\lambda_{\text{len}}}{2} + \lambda_{\text{len}}(b_i - 1/2).$$

The last term sums to zero over successes, so length comparisons redistribute the bonus around its fixed mean. This identity uses equal weights over complete trajectories, before expansion into turn or token records. The temperature controls preference sharpness; Figure 5 evaluates this choice, with 98.4% success at $\tau_{\text{len}} = 0.1$, compared with 93.8% at 0.05 and 96.1% at 0.2.

How does the efficiency bonus retain success priority? For the trajectory rewards in Equation 12, before invalid-action adjustments, a bonus restricted to successes preserves their priority under the following reward-ordering condition.

Corollary A.6 (Preservation of success–failure reward ordering). *Suppose the original rewards satisfy $R_i \geq R_j$ for every success i and failure j in a group, and $\lambda_{\text{len}} > 0$. Then the rewards in Equation 12 satisfy $R_i^{\text{pair}} > R_j^{\text{pair}}$ for every such pair. In particular, this holds for binary rewards $R_i = z_i$. Among successes with equal original rewards, the shaped rewards are strictly ordered by shorter length whenever their lengths differ.*

Proof. For every success, $b_i > 0$, including $b_i = 1/2$ when it is the only success. Hence

$$R_i^{\text{pair}} - R_j^{\text{pair}} = (R_i - R_j) + \lambda_{\text{len}} b_i > 0.$$

For two successes with equal original rewards, their shaped-reward difference is $\lambda_{\text{len}}(b_i - b_j)$, whose sign follows from Proposition A.5. □

The corollary establishes success priority at the reward-construction stage, while Proposition A.5 orders equally rewarded successes by efficiency. Table 2 reports 98.4% success for the full configuration, 93.0% without length shaping, and 97.7% with an absolute length bonus. Figure 6 reports the associated response-step curves. On unseen tasks, Figure 7 evaluates the complete method, with 89.6% success versus 66.4% for GRPO and reported mean turns of 10.06 versus 18.86. These experiments connect the reward's within-group preference to task completion and interaction efficiency at the policy level.

B Experimental Details

B.1 Training Configuration

SCOPE follows the supervised-initialization and online-training workflow of SEED [29]. Table 3 summarizes the key coefficients used by the full method. The policy snapshot supplies both the actor and the online analyzer; its skill-conditioned teacher scores are cached before student optimization. Online analysis has no separate supervised loss: the policy is updated through the joint objective in Equation 14.

Table 3: Key settings of full SCOPE. Parameter sweeps vary one of α, c, or τ_{len} at a time.

Parameter	Value
Maximum skill entries $K_{\max}$	5
Signed-gap slope α	2.5
Saturation bound c	1
Distillation weight λ_{OPSD}	0.01
Length-preference temperature τ_{len}	0.1
Length-bonus weight λ_{len}	5
PPO clipping radius ϵ	0.2
Reference KL weight β_{ref}	0.01
Entropy weight λ_H	0.001

The main comparison uses the step-150 checkpoint. ALFWorld ablations and parameter sweeps use step 180. Training-step savings in Figure 1(c) compare the first observed checkpoints reaching GRPO's final mean success rate. The ALFWorld SCOPE curve shows one run, while the GRPO and WebShop curves use three-seed means and sample standard deviations.

B.2 Supervised Data and Hindsight Analysis

GLM-5.2 generates the offline annotations used for SFT. Each input contains the task description, episode outcome, and the recorded sequence of observations and responses, together with action-validity feedback when available. Successful trajectories are analyzed for avoidable errors. Failed trajectories may additionally receive up to three successful action sequences from the same task as reference context. These offline references can come from separate runs of that task; online references are drawn from the current on-policy group.

The annotation configuration uses temperature 0, an output limit of 8,192 tokens, and disabled thinking. Observations are retained without character-level truncation. An annotation contains a trajectory summary and at most five skill entries, each with target-step indices, an error explanation, and a short reusable corrective instruction. Entries that require the same correction can cover multiple steps. The prompt asks the annotator to judge decisions using the information available at each step and to preserve reasonable exploration. An empty skill list is returned when no clear error is supported.

The data pipeline checks the JSON structure, nonempty required fields, and valid, unique target indices before export. Each SFT example is a system–user–assistant conversation: the input is the

analysis prompt, and the assistant target contains only the structured summary and skill entries. Successful reference action lists remain in the input rather than in the prediction target. The same analysis instructions and output schema are used by the online analyzer, which is the current policy snapshot. If a failed group contains no success, the reference block is omitted and the analyzer judges the recorded trajectory directly. Appendix C provides the templates.

B.3 Motivation Diagnostics

Outcome–step mismatch. The WebShop study in Figure 1(a) uses GLM-5.3 to identify erroneous steps in 1,400 trajectories: 536 successful and 864 failed episodes. Local decision quality is judged separately from episode outcome: an eventual success can contain avoidable mistakes, and an eventual failure can contain correct decisions. The figure reports error and correct-step proportions separately for the two outcome groups. GLM-5.2 supplies the offline SFT annotations described above; GLM-5.3 supplies these diagnostic error-step labels.

Off-target probability shifts. The skill-influence analysis scores 33,602 steps and 4,035,655 response tokens from 177 task groups. The recorded relevance labels divide steps into 14,081 related and 19,521 unrelated steps according to the applicability of the trajectory-level skill. On the sampled response tokens, we compute the signed gap $\Delta = \log \pi_T(y \mid h, s) - \log \pi_S(y \mid h)$. Within each outcome and relevance group, positive and negative bars are pooled-token means conditional on $\Delta > 0$ and $\Delta < 0$, respectively. Zero gaps are excluded from these conditional means. The 57.9% off-target share divides the number of nonzero-gap tokens at unrelated steps by the number of nonzero-gap tokens at all steps. The negative and positive shares of 53.9% and 46.1% use nonzero-gap tokens at unrelated steps as their denominator.

Error bars show 95% task-group bootstrap intervals from 2,000 resamples with random seed 0. Task descriptions and initial observations define the grouping proxy; each resample draws complete groups rather than individual tokens. This preserves within-group dependence when estimating uncertainty in the conditional means.

C SFT and Analyzer Prompt Templates

The offline annotator and online analyzer share the system message, analysis instructions, and JSON schema below. A user message consists of the appropriate outcome-specific opening, the shared instructions, and the episode context. Placeholders in angle brackets are replaced with the recorded task and trajectory. The reference block is included only for failures with available successful references. In the serialized trajectory, each step contains its index, observation, response, and action-validity flag when recorded.

System message

```
You analyze agent episodes and extract concise, policy-facing
skills for specific error steps. Return ONLY valid JSON.
```

Shared analysis instructions and output schema

```
First write a concise trajectory_summary. Then return up to 5
step_skills. For each:

1. Identify clear error steps in target_steps; steps requiring
the same correction share one skill.
2. Write one short policy-facing skill that extracts the
correction into a reusable workflow: the core decision rule and
action ordering that would avoid the mistake.
3. Write a one-sentence error_reason that cites the erroneous
steps and explains why the actions at those steps were mistakes.

Important constraints:
- Judge errors using information available at each target step;
do not label reasonable exploration.
- Write skill as one short, natural instruction, not a
retrospective explanation or a list of labeled sections.
- Generalize beyond this task without names, IDs, exact values,
or step numbers; keep the advice concrete and applicable at every
target step.
- Use existing step indices, sorted and unique; sort entries by
first target step.
- If no clear error is supported, still write trajectory_summary
and return {"step_skills":[]}.
- Return only the top-level fields trajectory_summary and
step_skills.

Return format (replace string placeholders and 0 with actual text
and existing integer step indices):
{"trajectory_summary":"string","step_skills":[{"target_steps":[0]
,"error_reason":"string","skill":"string"}]}
```

C.1 Successful Trajectories

The user message begins with the following opening, followed by the shared instructions and the episode context in the displayed order.

User template: successful trajectory

```
Analyze the following SUCCESSFUL trajectory. It may still contain
avoidable mistakes.
<SHARED ANALYSIS INSTRUCTIONS AND OUTPUT SCHEMA>

Episode context:
- Task description: <TASK>
- episode_success: success
- Interaction trajectory:
Step <STEP_INDEX>
Observation: <OBSERVATION>
Response: <RESPONSE>
Action valid: <true_or_false>

<REMAINING RECORDED STEPS>
```

C.2 Failed Trajectories

Successful reference trajectories contain the original action sequences. The opening explicitly asks the analyzer to assess the target trajectory rather than treating every difference from a successful reference as an error.

User template: failed trajectory

```
Analyze the following FAILED trajectory. Successful references
are hints; differences alone do not prove mistakes.
<SHARED ANALYSIS INSTRUCTIONS AND OUTPUT SCHEMA>

Episode context:
- Task description: <TASK>
- episode_success: failure
- Interaction trajectory:
Step <STEP_INDEX>
Observation: <OBSERVATION>
Response: <RESPONSE>
Action valid: <true_or_false>

<REMAINING RECORDED STEPS>
```

```
Successful reference trajectories (actions only):
Successful reference 1:
Action 0: <ACTION>
Action 1: <ACTION>
<REMAINING REFERENCE ACTIONS>

<ADDITIONAL SUCCESSFUL REFERENCES, IF AVAILABLE>
```